

Project Z — four system diagrams

Same Next.js product. Four views: website, Android app, Windows, and all together.

31 August 2026

One backend. Three shells. The website talks to the cloud on every click. Android and Windows can sell offline, then sync.

Diagram	What it is	Local data
1. Website	Browser on Vercel / localhost	IndexedDB fallback only; always prefers the network
2. Android app	Capacitor APK, same UI	Capacitor SQLite · 90-day invoice window
3. Windows	Tauri .exe, same UI	Tauri SQL plugin
4. All in one	How the three meet Turso, R2, mail, jobs	—

Shared path after the device: middleware → JWT → org header → /api/v1 → services → Prisma → Turso.

1. Website

Chrome / Edge on a laptop or shop PC. No APK, no .exe. The browser is only a thin client.

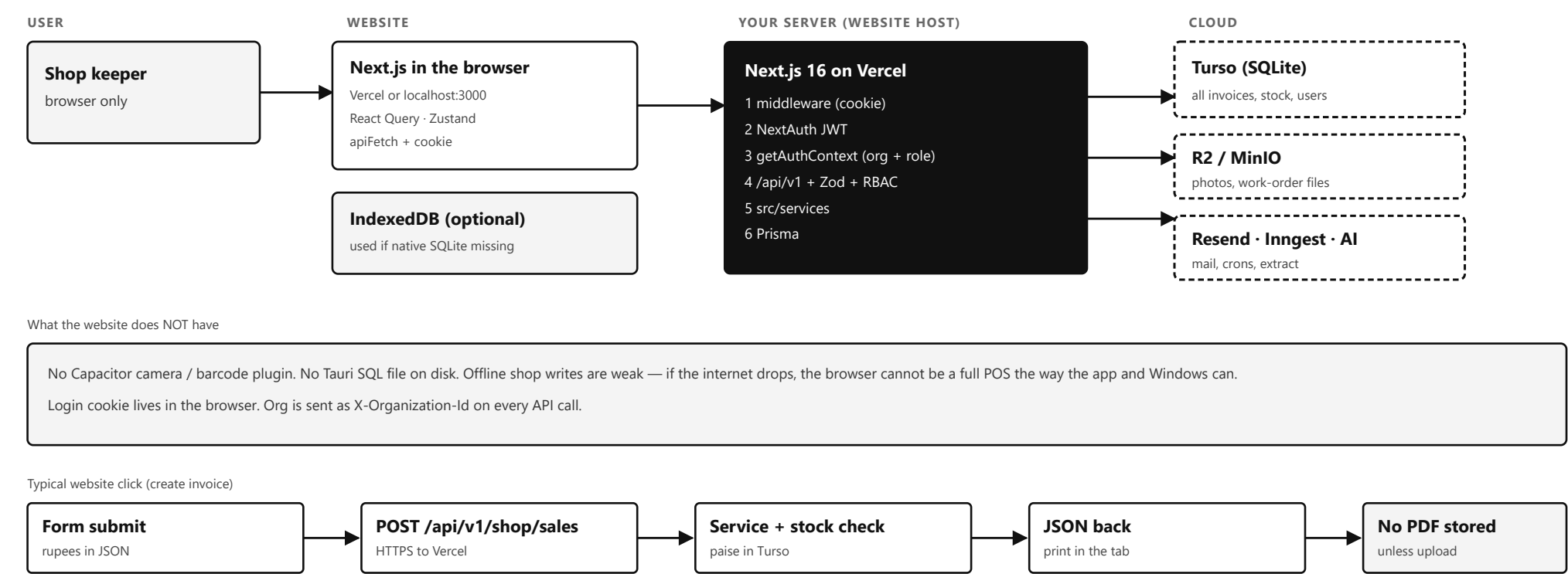


Figure 1 — Website. Online-first. Same APIs as the other two clients.

2. Android app

Capacitor wraps the same Next UI in an APK. Shop writes can go to the phone first, then sync when the network is back.

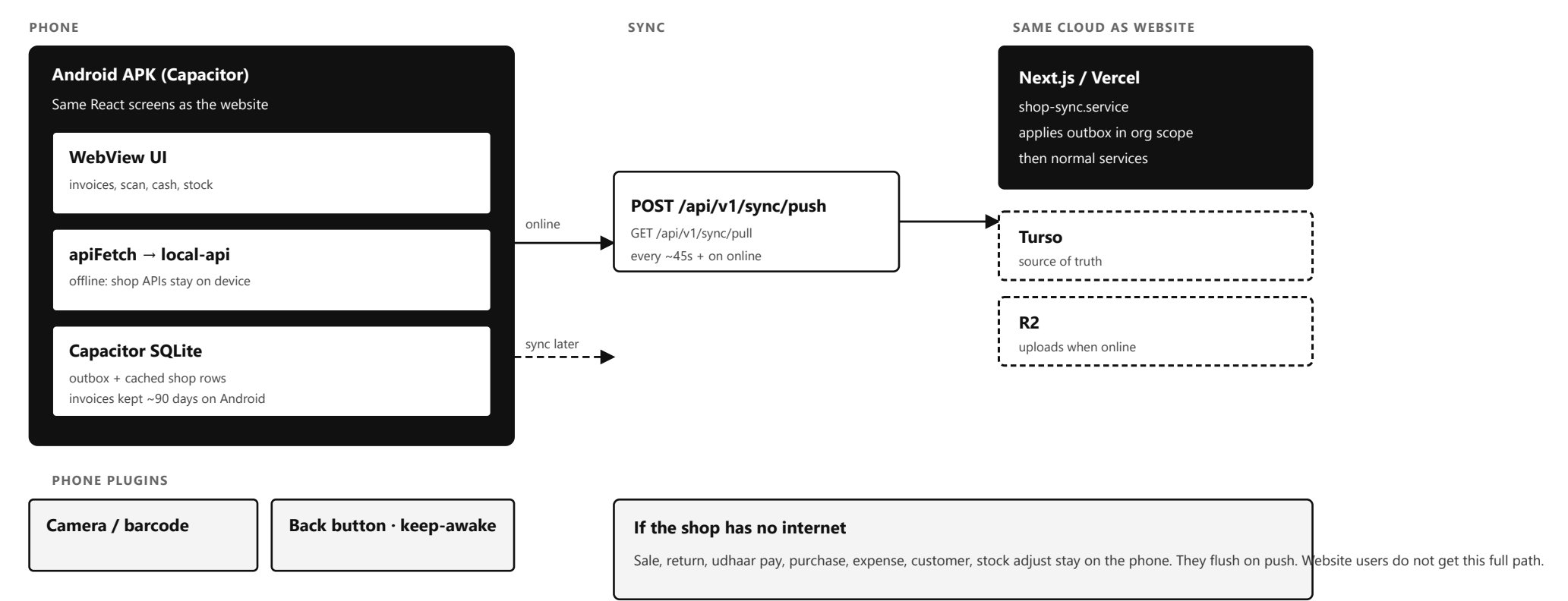


Figure 2 — Android. Device SQLite is a cache + outbox. Turso wins after sync.

3. Windows

Tauri .exe. Same React UI in a desktop WebView. Local SQL on the PC, not Capacitor SQLite.

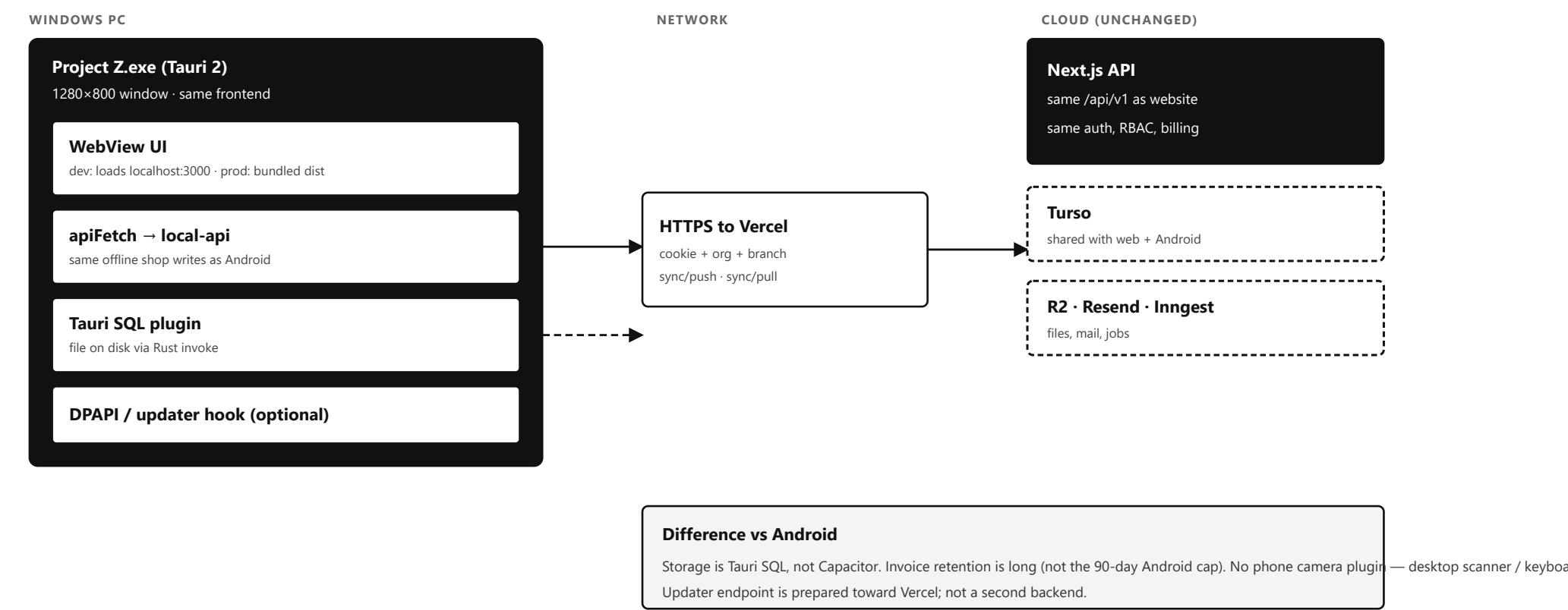


Figure 3 — Windows. Desktop shell + local SQL. Same cloud as the website and the phone.

4. All in one

Three clients. One API. One database. Files and jobs sit beside it, not inside Turso.

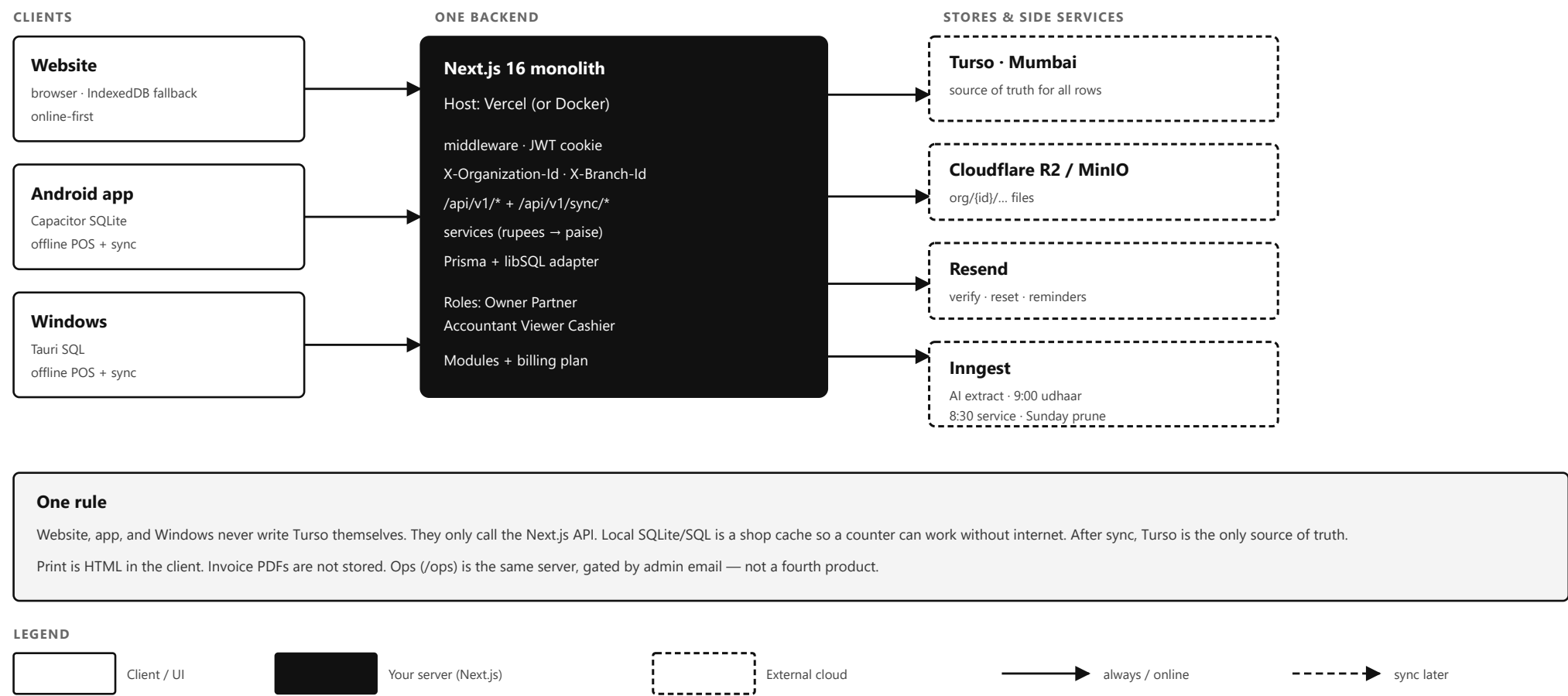


Figure 4 — All in one. Do not put photos in Turso. Do not give the phone a direct database URL.

How to read the four diagrams

	Website	Android app	Windows
Shell	Browser tab	Capacitor APK	Tauri exe
UI code	Same Next.js / React in src/app		
Local DB	IndexedDB fallback	Capacitor SQLite	Tauri SQL
Offline POS	Limited	Yes (shop routes)	Yes (shop routes)
Invoice cache	—	~90 days	Long
Device extras	—	Camera, back, keep-awake	Desktop window, DPAPI/updater
API	https://.../api/v1 on Vercel		
Truth	Turso after sync		

One request (all three, when online)

Client → cookie JWT → org + branch headers → getAuthContext → service → Prisma → Turso → JSON (BigInt as string).

Not in the pictures on purpose

Microservices (roadmap only). Postgres. Social login. SMS/push. Stored invoice PDFs. Builder vertical (removed).